HARSH GAJRA
Senior Software Engineer
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Over 10 years as a Staff/Senior Software Engineer, I specialize in building data-intensive, distributed applications across the full SDLC. I lead and mentor teams, consistently recognized for technical and soft skills, enabling effective cross-functional coordination and timely, high-quality project delivery. Seeking a Technology Lead/Engineering Manager role, I aim to apply my technical and leadership expertise to create resilient software and contribute to organizational success.

WORK EXPERIENCE

Senior Software Engineer – Crux Data (Feb 2023 – Present)

- Designed and implemented distributed, event-driven, scalable APIs for Crux's Data Platform using FastAPI, Pub/Sub, and PostgreSQL. These were containerized in Docker, orchestrated via Kubernetes, and deployed on GCP for high availability and reliability.
- Migrated over 2 billion Cassandra records to AlloyDB (PostgreSQL) using Python, Bash, SQL, Kubernetes, BigQuery and GCS, cutting annual infrastructure costs by over $400K.
- Architected Git-driven CI/CD framework automating complex data workflows via YAML, GitHub Actions, and merge queues, reducing pipeline deployment downtime by 1,800 hours annually.
- Led a team of 5 software engineers, providing technical guidance and mentorship to ensure successful project execution and delivery.
- Lead Engineer for cruxctl, a Python CLI tool built with the Typer framework, providing customers with command-line access to Crux's External data platform.
- Accelerated onboarding and reduced manual tuning of Apache Airflow ingestion pipelines by developing an AI-powered adapter recommendation engine using OpenAI GPT LLMs to automate configuration.
- Designed, implemented, and deployed scalable ETL pipelines leveraging pandas for data manipulation and analysis, orchestrating workflows with Apache Airflow DAGs, and utilizing Google BigQuery as an OLAP data warehouse for high-performance querying.
- Developed and deployed asynchronous Java APIs using Spring Boot, PostgreSQL, and Kafka on Kubernetes in GCP for reliable messaging and microservice decoupling.

Software Engineer - Amazon Web Services, Data Migration Service
Seattle, Washington (Sept,2021 – Feb 2023)

- Designed and developed a distributed test execution platform leveraging serverless technologies (AWS Lambda, AWS Simple Workflow, AWS CloudWatch, Cloud Development Kit) and Java 11. This platform manages and executes hundreds of thousands of parallel test cases for various AWS Data Migration Service components, including Control Plane, Data Plane (Kamino), and Serverless, enabling testing at scale.
- Worked on re-architecting the Canary and Test infrastructure to eliminate deployment bottlenecks and congestions leading to overall stabilization and faster deployments saving 50 hours a week.
- Started as 1st hire on the Platform STP team, mentored and led a team of 3 interns and 2 engineers, and served as subject matter expert for multiple components of the software product.
- Wrote CI/CD pipelines using AWS Cloud development kit and CloudFormation to perform infrastructure deployments predictably and repeatedly, with rollback on error.
- Wrote automation, identified and eliminated bugs, fixed design flaws across the product stack concerning AWS DMS bringing down on call support tickets by 60 % and saving 2400 hours yearly.

Software Developer 2 – FINRA
Reston, Virginia (May 2020 - Sept, 2021)

- Designed and developed the core components of FINRA’s Consolidated Audit Cat (CAT) platform, enabling the capture and traceability of billions of transactions across multiple U.S. stock exchanges to meet SEC regulatory mandates.
- Developed and optimized advanced SQL queries to process and analyze high-volume market data, producing actionable reports and insights that enhanced project dashboards used by leadership for operational decision-making.

Software Engineer - Pype Inc (Acquired by AutoDesk in May 2020)
Herndon, VA, USA (July 2018 – May 2020)

- Architected and developed the backend for Pype's Autospecs and eBinder software products using Java 8, Spring Boot, JPA, Redis and MySQL databases.
- Designed and developed software solutions for optimizing the business process followed by in-house sales and support teams thus saving the organization $360,000 per year in costs and 15,000 man-hours per year.
- Designed applications using the message brokers like RabbitMQ for message queuing and asynchronous job processing.
- Algorithm Development: Design, Analyze and implement algorithms in Java for submittal information extraction from construction drawing sheets.
- Designed, analyzed, and implemented algorithms for extracting submittal information from construction drawing sheets.

TECHNICAL SKILLS

Programming Skills: Python 3, Java, Bash, FastAPI, Pub/Sub, Kafka, Typer, AWS, Spring Boot.
Databases: Amazon RDS, Postgres, AlloyDB, Cassandra, Amazon Aurora, MySQL, DynamoDB, BigQuery.
Tools: GitHub, JIRA, Apache Airflow, GPT, LLMs, AI, GitHub Actions, Docker, Kubernetes, Co-Pilot, Cursor.
Other: Brazil Build System, AWS CDK, SWF, AWS, Cloudwatch, EC2, GCP, CodeFresh CI/CD.

EDUCATION

Masters in Computer Science - State University Of New York, Binghamton, August 2016 - May 2018
Bachelor of Computer Engineering - K.J.Somaiya College of Engineering, Mumbai, First class (2011- 2015)

PAPER PUBLICATIONS

Paper: Credit card Fraud Detection and Prevention Using Perceptron Training Algorithm.
Journal: International Journal on Recent and Innovation Trends in Computing and Communication, Apr 2015
Link: https://ijritcc.org/index.php/ijritcc/article/view/4182/4182